CLASS 9 — SCIENCE CHAPTER TEST SET A

Chapter 6: **How Forces Affect Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The net force needed to give a body of mass 5 kg an acceleration of 4 m s^{-2} is: 1

(a) 40 N **(b)** 20 N **(c)** 1.25 N **(d)** 9 N

2. A net force of 20 N acts on a body of mass 4 kg. Its acceleration is: 1

(a) 0.2 m s^{-2} **(b)** 80 m s^{-2} **(c)** 10 m s^{-2} **(d)** 5 m s^{-2}

3. The weight of a 5 kg body is (g = 10 m s^{-2}): 1

(a) 50 N **(b)** 25 N **(c)** 5 N **(d)** 100 N

4. Two forces of 8 N and 8 N act on a block along the same straight line in opposite directions. The magnitude of the net force is: 1

(a) 16 N **(b)** 0 N **(c)** 8 N **(d)** 64 N

5. A force of 45 N produces an acceleration of 9 m s^{-2} in a body. Its mass is: 1

(a) 9 kg **(b)** 5 kg **(c)** 405 kg **(d)** 10 kg

6. 1 N is the force that gives an acceleration of 1 m s^{-2} to a mass of: 1

(a) 1 g **(b)** 10 kg **(c)** 1 kg **(d)** 9.8 kg

7. A body moving at constant velocity has net force: 1

(a) increasing **(b)** zero
(c) equal to mv **(d)** equal to its weight

8. A fielder pulls his hands back while catching a fast ball to: 1

(a) increase the ball's mass
(b) decrease the ball's momentum to zero faster
(c) increase the time of stopping and reduce the force
(d) increase the force

9. A rocket lifts off because the exhaust gases pushed downwards exert an equal and opposite force which is: 1

(a) zero
(b) upward on the rocket and greater than its weight
(c) upward but less than its weight
(d) downward on the rocket

10. Two teams in a tug of war pull with 500 N each and the rope does not move. The forces on the rope are: 1

(a) action–reaction on the rope **(b)** balanced

(c) unbalanced **(d)** zero on each team

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. State Newton's first law of motion. 2

12. What is inertia? On what does it depend? 2

13. Why does a passenger lurch forward when a moving bus stops suddenly? 2

14. State Newton's second law and write its mathematical form. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Describe an activity to show that friction depends on the nature of surfaces. 3

16. Explain Galileo's thought experiment that led to the first law. 3

17. Does a body moving with constant velocity have a net force on it? Explain with the example of a box pushed on a floor. 3

OR

Draw the position–time and velocity–time graphs for a body with no net force and explain. 3

18. A body of mass 10 kg is acted upon by a force of 50 N. Find the acceleration and the distance covered in 4 s from rest. 3

OR

A car of mass 1000 kg moving at 20 m s^{-1} is stopped in 5 s. Find the force applied by brakes. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) State Newton's first law. 5

(ii) Describe Galileo's reasoning.

(iii) Give two everyday examples of inertia.

OR

(i) Describe Activity 6.3 to show that acceleration depends on force. 5

(ii) How does mass affect acceleration?

(iii) Write the law in the mathematical form.

20. **Case study:** A student pushes a 25 kg block; maximum friction is 50 N. 5

(i) What happens with a 50 N push?

(ii) What is the net force with 55 N?

(iii) Find the acceleration.

(iv) Find its displacement in 2 s.

OR

(i) State Newton's third law. 5

(ii) Explain with the example of a canoe.

(iii) Why do the forces not cancel?

END OF QUESTION PAPER — Total: 40 Marks